BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Autumn Assessment (AP1)

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Which integers can random(1,6) produce in OCR ERL?

2. Why use pseudocode?

A) Run directly on the CPU
B) Replace all program testing
C) Draw the user interface
D) Plan steps without exact syntax

3. "A diamond is used only for unconditional calculations." Correct this. Explain why.

4. A program works out a bill, then applies a discount. "The order of steps never matters." Correct this. What could go wrong?

Today's objectives

Recall key knowledge independently before the assessment. Show accurate working on calculations, pseudocode and explanations.

Key vocabulary - Read, cover, recall

Method marks
Credit for correct working even if a final numerical result is wrong.

Trace
A record of the executed operations and changing variable values.

Command word
An instruction such as state, explain, calculate or evaluate.

Assessment
An independent measure of current knowledge and skills.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Read the question	Underline the command word and identify the information provided.
Allocate the marks	For explain and compare questions, include distinct relevant causal points.
Show your working	Give calculation steps with units; record indexes, tests and intermediate values.
Check answers	Review the final value, boundaries, condition logic and terminology.

I Do - One worked example

A separate revision example: calculate the uncompressed size of a 10 x 20 bitmap with 4-bit colour depth, in bytes.

1. Write width=10, height=20, colour depth=4 bits.
2. Calculate 10 x 20 = 200 pixels.
3. Calculate 200 x 4 = 800 bits.
4. Convert 800 / 8 = 100 bytes. This example is not from AP1.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Trace total=0; for i=1 to 2; total=total+i; next i. Give both updates and the output.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the output. Then explain the variable's change.

```
value = 7
value = value + 2
print(value)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Explain why showing steps can earn marks even if a final calculation is wrong. [2]

Q3. Explain why an answer should use the question's command word. [2]

Q4. State one reason to check the output of a trace table. [1]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Show calculation steps and state units where a question needs them. ☐ T ☐ F

2. Underline the command word and identify the information provided. ☐ T ☐ F

3. For explain and compare questions, include distinct relevant causal points. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Any integer from 1 through 6 inclusive. ☐

Q2 D - Plan steps without exact syntax ☐

Q3 A decision or condition with alternative paths. ☐

Q4 Steps must be clear and in the right order. Check that conditions give the right result for every allowed input. ☐

Code First - Model

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We Do - Check the model

1. Final printed value is 3. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Correct method can earn method credit [1]; a visible step enables identification of what was done accurately [1]. ☐

Q3 State/name and explain/evaluate require different levels of detail [1]; matching the command focuses on assessable evidence [1]. ☐

Q4 Checking that the final output agrees with executed variable updates [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: